

TRACK 1 — AI & INTELLIGENT SOLUTIONS

Leverage ML and AI to improve social outcomes, optimise resource distribution, and democratise information access.

Challenge 1.1 — AI-Based Multilingual Chatbot for Welfare Scheme Awareness

Domain / Track: AI & Intelligent Systems

Background & Context India runs over 950 central and state welfare schemes — PM-KISAN, Ayushman Bharat, PMAY, Ujjwala, NREGA, Sukanya Samriddhi, and hundreds of state-level entitlements. Yet a 2023 NITI Aayog field study estimated that fewer than 40% of eligible rural beneficiaries are aware of the schemes they qualify for, and an even smaller share complete the application successfully. The bottlenecks are not benefits design — they are information asymmetry, language barriers, low literacy, and intimidating documentation. Most government portals are English/Hindi-only, assume smartphone literacy, and break under low bandwidth.

Problem Statement Design a low-bandwidth, multilingual conversational assistant — deployable over WhatsApp, SMS, or IVR — that helps a rural user discover schemes they're eligible for, understand the documents required, and walk through the application process in their own language and dialect.

Why This Matters A single successful PM-KISAN enrolment is ₹6,000/year for a farming household; a successful Ayushman Bharat enrolment can prevent a medical-debt spiral. Closing the awareness gap is among the highest-leverage social interventions in India today — no new policy needed, only better last-mile delivery.

Project Objectives

- Build a working chatbot supporting at least two regional languages (e.g., Hindi + Tamil, or Bengali + Marathi).
- Cover a focused catalogue of 8–10 high-impact schemes rather than all 950.
- Design an eligibility-questioning flow that asks 4–6 simple questions and returns a personalised shortlist.
- Generate a downloadable/SMS-able checklist of required documents in the user's language.

Key Questions to Explore

- What is the lowest-friction channel for a user with a ₹500 keypad phone — WhatsApp, SMS, or IVR?

- How do you handle code-mixing (e.g., "मेरा aadhaar खो गया")?
- How do you keep the model honest when it doesn't know the answer, rather than hallucinating eligibility?
- What is the offline fallback when the user has no data balance?

Suggested Approach

- Use an LLM (Gemini, Llama 3, or an open Indic model like Sarvam-1 / IndicTrans2) behind a retrieval layer grounded in the official scheme database scraped from MyScheme.gov.in.
- Use RAG (retrieval-augmented generation) to prevent hallucinated entitlements — answers must cite the scheme.
- Build the front-end on Twilio WhatsApp Sandbox or Meta WhatsApp Business API (free tier).
- Compress the conversation flow into 4–6 turn dialogs to respect low bandwidth.

Recommended Tools & Resources

- APIs/Frameworks: Twilio, WhatsApp Business Cloud API, Rasa, LangChain, Hugging Face Transformers.
- Indic NLP: AI4Bharat IndicTrans2, IndicBERT, Bhashini.
- Datasets: MyScheme.gov.in scrape, data.gov.in scheme catalogues.
- Deployment: Replit, Render, or Hugging Face Spaces (free tiers).

Expected Deliverables

- Functional chatbot prototype (live demo link or video).
- User-flow diagram covering at least 3 personas (farmer, gig worker, woman head-of-household).
- Pilot test report with 10–15 real users from any nearby village or community.
- 2-page impact projection.

Success Metrics

- $\geq 80\%$ of test users complete the eligibility flow without dropping off.
- $\geq 70\%$ comprehension score on scheme summaries (measured via post-chat quiz).
- < 3 seconds median response time on 2G/3G simulation.

Potential Impact If adopted by a single district administration or NGO with 10,000 beneficiaries, even a 20% lift in scheme uptake translates to crores of rupees in entitlements reaching households that currently miss out.

Challenge 1.2 — Predictive Analytics for Resource Allocation in Disaster & NGO Operations

Domain / Track: AI & Intelligent Systems

Background & Context India faces recurring monsoon floods (Assam, Bihar, Kerala), cyclones (Odisha, Andhra, West Bengal), heatwaves (now a notified disaster), and drought cycles. NGOs and district disaster authorities still rely heavily on reactive logistics — supplies arrive after media coverage peaks. Meanwhile, NDMA, IMD, and ISRO release rich open datasets that are rarely fused at the operational layer.

Problem Statement Build a machine-learning model that ingests historical demographic, climatic, and incident data to forecast where and how much of a critical resource (medical kits, dry rations, tarpaulin, ORS, drinking water) will be needed over the next 7–30 days in a chosen district or NGO operating area.

Why This Matters Pre-positioning supplies 48 hours before a flood event reduces relief-delivery cost by ~30% and saves lives. For NGOs running on slim budgets, predictive allocation is the difference between serving 5,000 or 15,000 people with the same money.

Project Objectives

- Choose ONE focused use case (e.g., flood-relief kit demand in coastal Odisha, OR maternal-health supply demand in Jharkhand PHCs).
- Build a forecasting model with documented accuracy metrics (MAE, RMSE).
- Ship an interactive dashboard a non-technical NGO ops manager can use.

Key Questions to Explore

- Which features matter most — rainfall, river-gauge data, population density, road accessibility, or past incident frequency?
- How do you handle data gaps in remote districts?
- How do you communicate model uncertainty to a relief coordinator who needs a yes/no decision?

Suggested Approach

- Time-series models: Prophet, ARIMA, or LightGBM with lag features for tabular forecasting.
- Geospatial layer: GeoPandas + Folium for district-level visualisation.
- Stretch goal: integrate IMD rainfall forecast API for forward-looking signals.

Recommended Tools & Resources

- Datasets: data.gov.in, NDMA incident database, IMD, Bhuvan (ISRO), OpenStreetMap, SEDAC population grids.
- Stack: Python, Scikit-learn, Prophet, GeoPandas, Streamlit/Dash for dashboard.
- Reference: SPHERE Handbook (humanitarian standards) for per-capita supply norms.

Expected Deliverables

- Trained model with reproducible notebook and accuracy report.
- Deployment-ready Streamlit/Dash dashboard.
- 1-page resource-allocation playbook describing how an NGO would operationalise the forecasts.

Success Metrics

- Forecast MAPE < 20% on hold-out historical data.
- Dashboard loads under 5 seconds; usable by a non-technical user without training.
- At least one NGO contact validates the output as actionable.

Potential Impact Adoption by even a single district disaster cell or a state-level NGO network could shave days off relief response and cut wastage of perishable supplies.

Challenge 1.3 — Intelligent Systems for Public Service Access

Domain / Track: AI & Intelligent Systems

Background & Context India has 2.68 crore people with disabilities (Census 2011 — almost certainly an undercount), yet most public spaces — bus stops, ration shops, PHCs, banks — remain functionally inaccessible. Separately, district hospitals and PHCs routinely see 4–6 hour wait times for OPD consultations, with patients (often daily-wage earners) losing entire workdays. Computer vision and IoT have matured to the point where simple, low-cost interventions are feasible by an undergraduate team.

Problem Statement Pick ONE of the following:

- **Track A — Accessibility:** Build a CV/IoT system that improves a specific public service for differently-abled users (e.g., a smartphone app that reads signboards aloud in regional languages, an audio-guidance beacon for a PHC, a queue-position announcer for the visually impaired).
- **Track B — Queue Management:** Build an intelligent token/queue system for a high-footfall PHC or government office that estimates wait times, allows remote check-in over SMS/WhatsApp, and reduces in-clinic crowding.

Why This Matters A 30-minute reduction in average PHC wait time across India's ~30,000 PHCs would return tens of millions of productive hours annually. Accessibility upgrades convert "officially available" services into "actually usable" services for crores of citizens.

Project Objectives

- Build a working prototype (hardware mock-up or pure-software is acceptable).

- Conduct a short accessibility/usability audit with at least 5 representative users.
- Document the unit economics — what would it cost to deploy across 10 PHCs?

Key Questions to Explore

- What is the cheapest hardware footprint that delivers 80% of the value?
- How do you design for users who may not own a smartphone?
- Where does the system gracefully degrade — power cut, no internet, untrained staff?

Suggested Approach

- Vision/audio: OpenCV, YOLOv8-nano, MediaPipe, Whisper for STT, Coqui/Bhashini for Indic TTS.
- Queue system: Twilio SMS + simple FastAPI backend + a tablet dashboard for the clinic.
- Hardware (optional): Raspberry Pi Zero W (~₹2,500) or an old Android phone as the edge device.

Recommended Tools & Resources

- OpenCV, YOLO, MediaPipe, Whisper, Bhashini TTS, Twilio, FastAPI, Firebase.
- WHO accessibility guidelines, Rights of Persons with Disabilities Act (2016) compliance checklist.

Expected Deliverables

- Working prototype or detailed technical blueprint with BOM (bill of materials).
- Accessibility/usability audit report.
- Estimated throughput improvement (queue) or task-completion lift (accessibility).

Success Metrics

- Measurable reduction in task time or wait time vs. baseline (target $\geq 30\%$).
- Per-deployment cost under ₹10,000.
- Positive usability rating from $\geq 80\%$ of test users.

Potential Impact A replicable kit that any panchayat, NGO, or hospital admin could deploy without specialised engineering support.